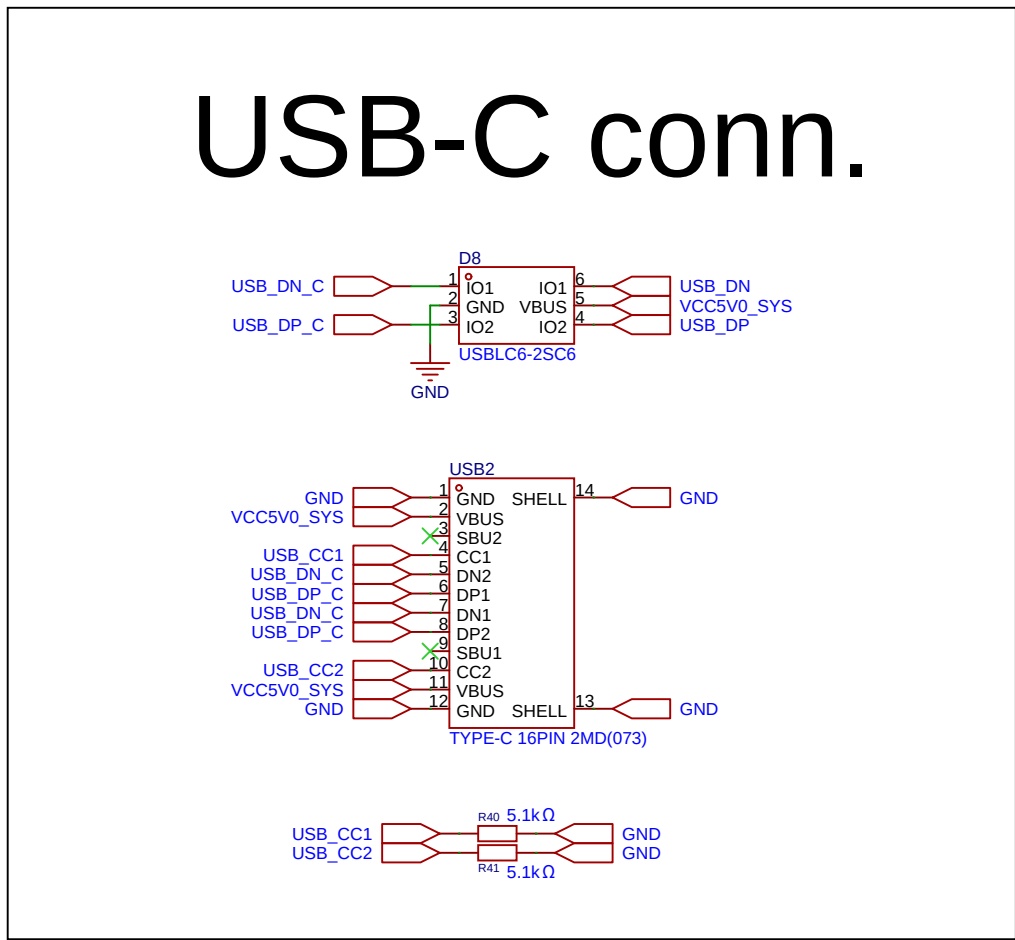


3V3 LDO + LED



USB-C conn.

The diagram illustrates the pin connections for a USB-C connector. The connector is labeled "D8" and "USBLC6-25C6". The pins are connected as follows:

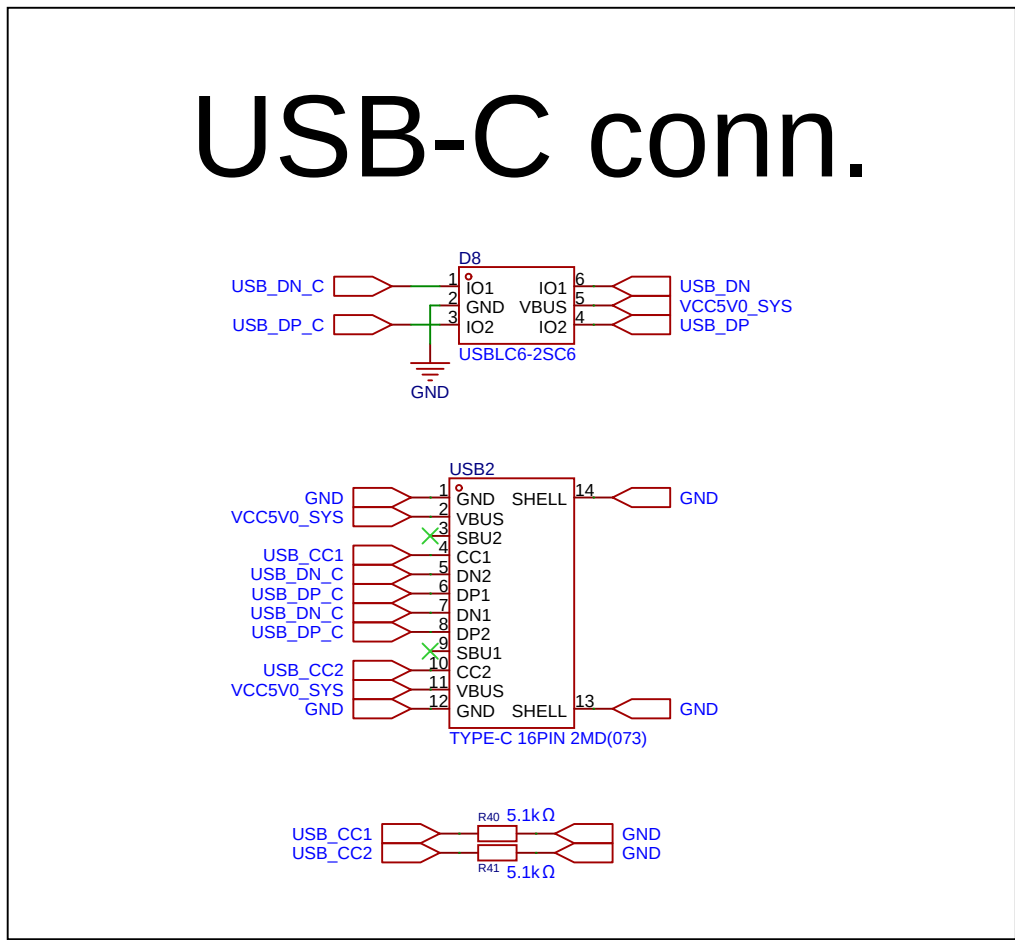
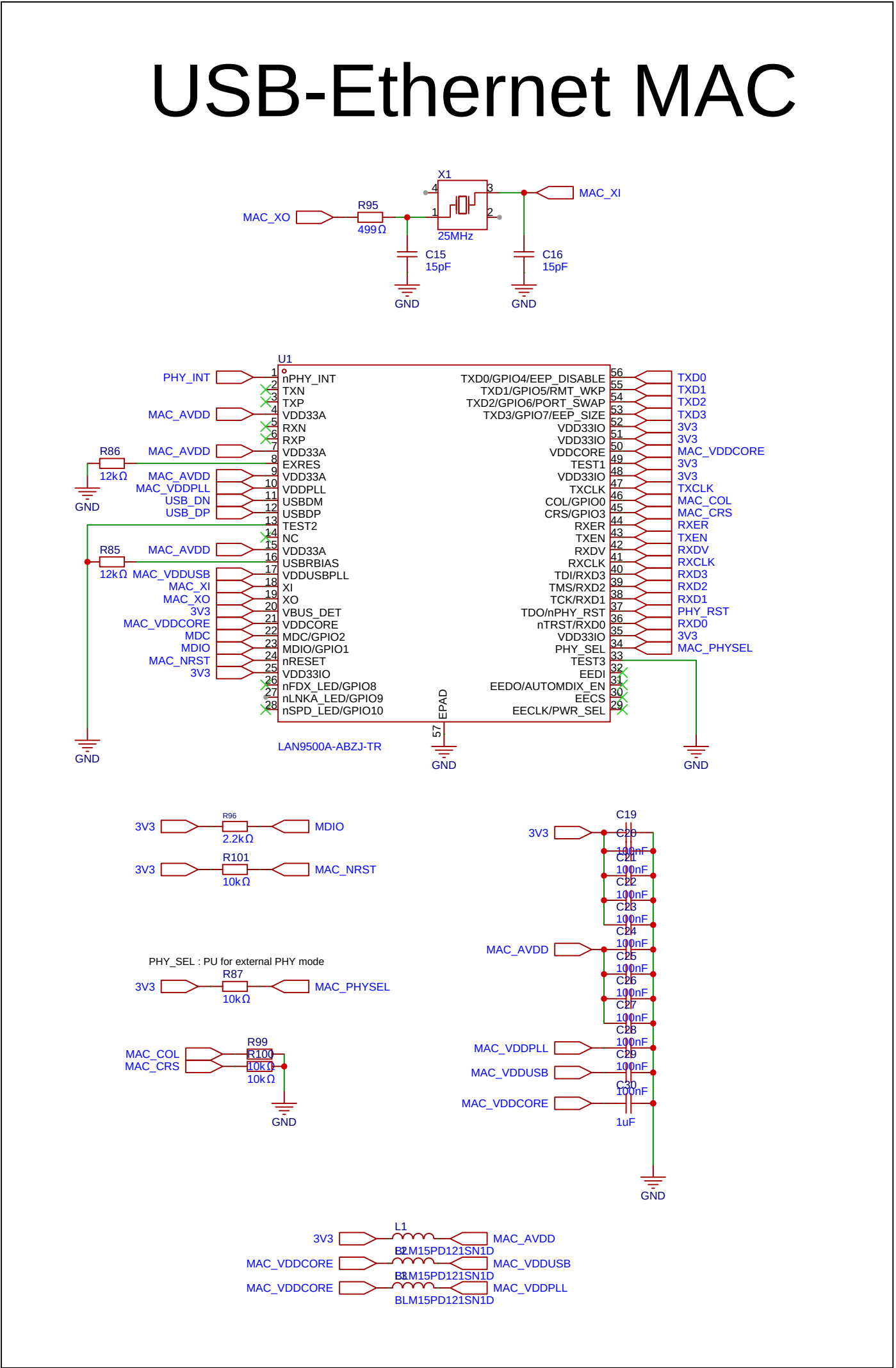
- Pin 1:** USB_DN_C
- Pin 2:** USB_DP_C
- Pin 3:** GND
- Pin 4:** GND
- Pin 5:** USB_DN
- Pin 6:** VCC5V0_SYS
- Pin 7:** USB_DP
- Pin 8:** GND
- Pin 9:** GND
- Pin 10:** GND
- Pin 11:** GND
- Pin 12:** GND
- Pin 13:** GND
- Pin 14:** GND
- Pin 15:** GND
- Pin 16:** GND

The USB2 connector is also shown, with the following connections:

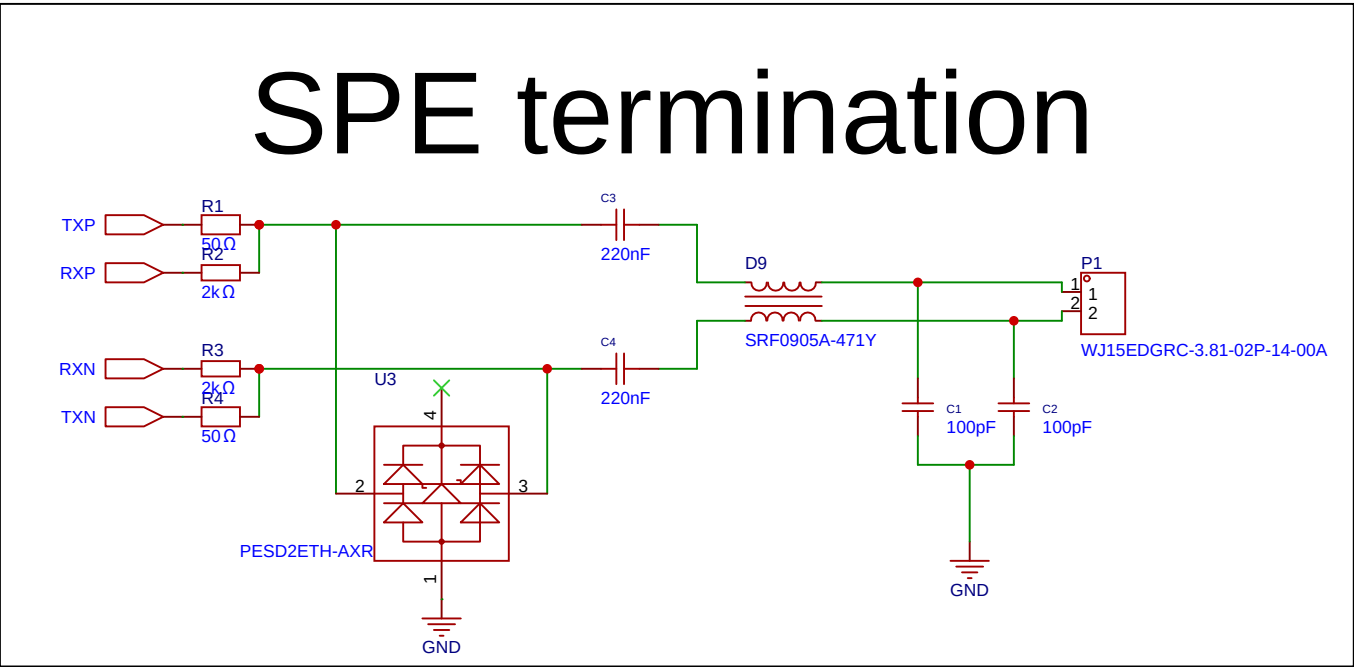
- Pin 1:** GND
- Pin 2:** VCC5V0_SYS
- Pin 3:** SBU2
- Pin 4:** CC1
- Pin 5:** DN2
- Pin 6:** DP1
- Pin 7:** DN1
- Pin 8:** DP2
- Pin 9:** SBU1
- Pin 10:** CC2
- Pin 11:** VBUS
- Pin 12:** GND
- Pin 13:** GND
- Pin 14:** GND
- Pin 15:** GND
- Pin 16:** GND

The USB2 connector is labeled "TYPE-C 16PIN 2MD(073)".

The diagram also shows the connection of the USB2 connector to the USB-C connector via a 5.1kΩ resistor (R40) and a 5.1kΩ resistor (R41).

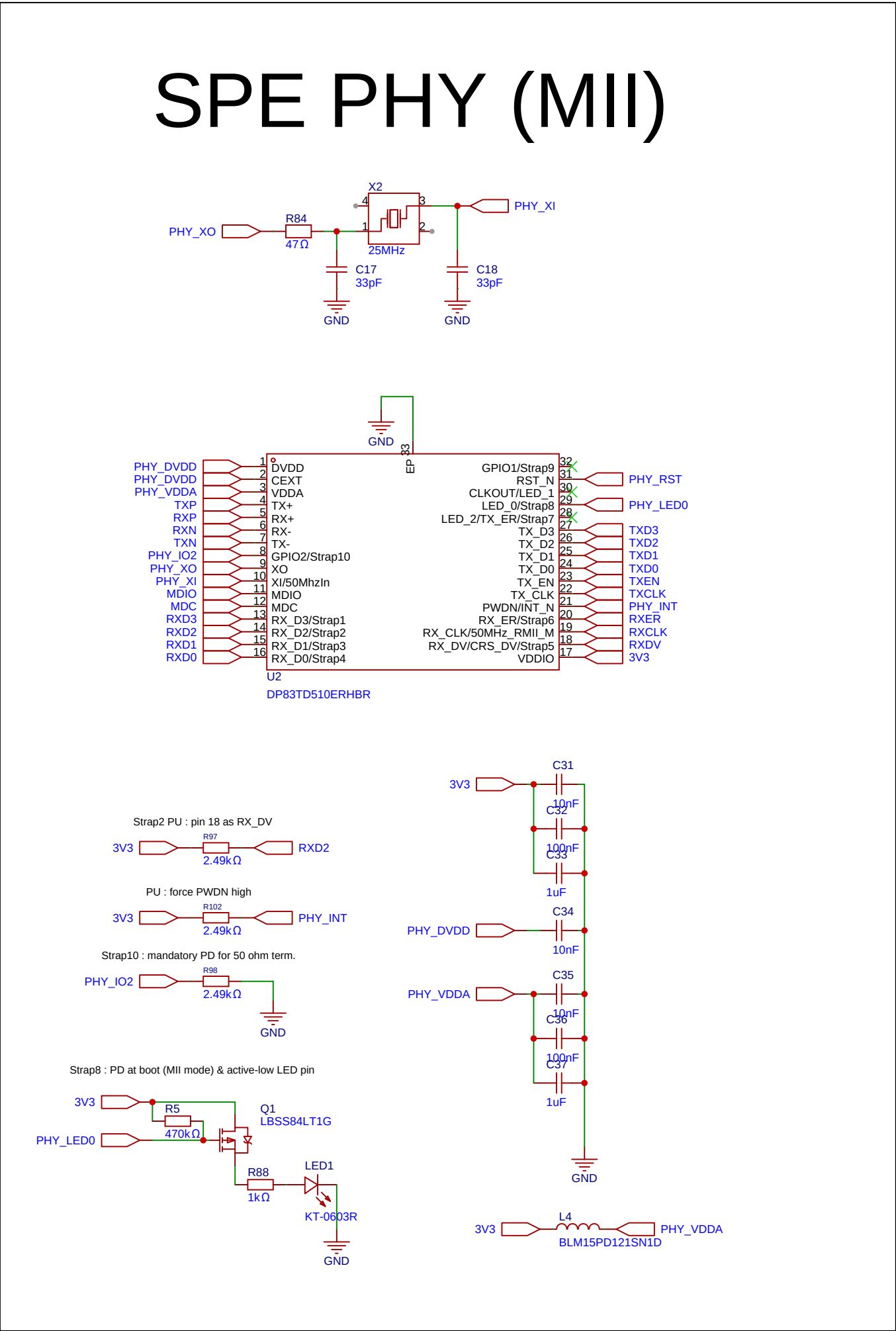
[illegible]

SPE termination



SPE PHY (MII)

The schematic diagram illustrates the SPE PHY (MII) circuit. The top section shows the PHY_XO crystal oscillator circuit, which includes a 25MHz crystal (X2), a 47Ω resistor (R84), and capacitors C17 (33pF) and C18 (33pF) connected to ground. The bottom section shows the DP83TD510ERHBR (U2) chip with its pin connections. On the left, PHY_DVDD, PHY_DVDD, PHY_VDDA, TXP, RXN, TXN, PHY_IO2, PHY_XO, PHY_XI, MDIO, MDC, RXD3, RXD2, RXD1, and RXD0 are connected to the chip's pins. On the right, GPIO1/Strap9, RST_N, CLKOUT/LED_1, LED_D/Strap8, LED_2/TX_ER/Strap7, TX_D3, TX_D2, TX_D1, TX_D0, TX_EN, TX_CLK, PWDN/INT_N, RX_ER/Strap6, RX_CLK/50MHz_RMII_M, RX_DV/CRS_DV/Strap5, and VDDIO are connected to the chip's pins. The bottom section shows the DP83TD510ERHBR (U2) chip with its pin connections. On the left, PHY_DVDD, PHY_DVDD, PHY_VDDA, TXP, RXN, TXN, PHY_IO2, PHY_XO, PHY_XI, MDIO, MDC, RXD3, RXD2, RXD1, and RXD0 are connected to the chip's pins. On the right, GPIO1/Strap9, RST_N, CLKOUT/LED_1, LED_D/Strap8, LED_2/TX_ER/Strap7, TX_D3, TX_D2, TX_D1, TX_D0, TX_EN, TX_CLK, PWDN/INT_N, RX_ER/Strap6, RX_CLK/50MHz_RMII_M, RX_DV/CRS_DV/Strap5, and VDDIO are connected to the chip's pins.



Schematic	Schematic1			Update Date	2026-06-06
				Create Date	2026-05-28
Page	P1			Part Number	
Drawn		spe-T1L_usb_adapter			
Reviewed					
		VER	SIZE	PAGE	1 OF 2
		V1.1	A4	EasyEDA.com	

Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO		
DVDD	1	A	Digital supply 1.0 V • For single-supply operation: Short this pin with CEXT (Pin 2) • Optional (dual-supply operation): Connect external 1.0 V to achieve lowest power Refer to Power Connection Diagram in Application section
CEXT	2	A	External capacitor for internal LDO • For single-supply operation: Connect 0.01- μ F capacitor and short it with pin 1 • For dual-supply operation, leave unconnected Refer to Power Connection Diagram in Application section
VDDA	3	A	Supply 3.3 V to support both 2.4-V p2p and 1-V p2p mode. Supply 1.8 V to support only 1-V p2p mode. Supplied voltage will be reflected in bit 13 of auto negotiation base page as capability to support 2.4-V p2p or 1-V p2p. 0x20E, bit 13 = 1 when 3.3 V is selected. 0x20E, bit 13 = 0 when 1.8 V is selected. Ensure the Strap7 "Reach Selection" strap is selected appropriately to request the output voltage level in the auto negotiation page.
TX+	4	A	TX+, TX- : Differential Transmit Output (PMD): These differential outputs are configured to 2.4-V p2p or 1-V p2p mode based on configuration chosen for PHY and auto negotiation with Link Partner.
RX+	5	A	RX+, RX- : These differential inputs are automatically configured to accept 2.4-V p2p or 1-V p2p mode based on configuration chosen for PHY.
RX-	6	A	
TX-	7	A	TX+, TX- : Differential Transmit Output (PMD): These differential outputs are configured to 2.4-V p2p or 1-V p2p mode based on configuration chosen for PHY and auto negotiation with Link Partner.
GPIO2	8	Strap	GPIO: This pin can be configured for multiple configuration thru register configuration. It has mandatory PU or PD strap. Refer to Straps sections for details.
XO	9	A	Crystal Output: Reference Clock output. XO pin is used for crystal only. This pin should be left floating when a CMOS-level oscillator is connected to XI.
XI/50MHzIn	10	A	Crystal / Oscillator Input Clock MII, RMII master mode: 25-MHz $\pm$ 50 ppm-tolerance crystal or oscillator clock RMII slave mode: 50-MHz $\pm$ 50 ppm-tolerance CMOS-level oscillator clock
MDIO	11		Management Data I/O: Bi-directional management data signal that may be source by the management station or the PHY. This pin requires an external pull of 2.2k Ω - 4.0 k Ω .
MDC	12		Management Data Clock: Synchronous clock to the MDIO serial management input/output data. This clock may be asynchronous to the MAC transmit and receive clocks. The maximum clock rate is 1.75 MHz.
RX_D3	13	Strap	Receive Data: Symbols received on the cable are decoded and presented on these pins synchronous to the rising edge of RX_CLK. They contain valid data when RX_DV is asserted. A nibble RX_D[3:0] is received in MII mode. 2-bits RX_D[1:0] is received in RMII mode.
RX_D2	14	Strap	
RX_D1	15	Strap	
RX_D0	16	Strap	
VDDIO	17	Power	I/O Supply : 3.3 V/2.5 V/1.8 V. For decoupling capacitor requirements , refer to Power Connection Diagram in Application section.
RX_DV/ CRS_DV	18	Strap	Receive Data Valid: This pin indicates valid data is present on the RX_D[3:0] for MII mode and on RX_D[1:0] for RMII mode. In RMII mode, this pin acts as CRS_DV and combines the RMII arrier and Receive Data Valid indications. This pin can be configured to RX_DV to enable RMII repeater mode using strap or register configuration. RGMII mode: RGMII Receive Control: RX_CTRL combines receive data valid and receive error signals. RX_DV is presented on the rising edge of RX_CLK and RX_ER on the falling edge of RX_CLK.

PIN		TYPE	DESCRIPTION
NAME	NO		
RX_CLK/ 50MHz_RMII_M	19		MIl Receive Clock: MIl Receive Clock provides a 2.5-MHz reference clock for 10-Mbps speed, which is derived from the received data stream. In RMII master mode, this provides 50-MHz reference clock. In RMII slave mode, this pin is not used and remains Input/PD. RGMII Receive Clock: RGMII Receive Clock provides a 2.5-MHz reference clock for 10-Mbps speed, which is derived from the receive data stream.
RX_ER	20	Strap	Receive Error: This pin indicates that an error symbol has been detected within a received packet in both MII and RMII mode. In MII mode, RX_ER is asserted high synchronously to the rising edge of RX_CLK. In RMII mode, RX_ER is asserted high synchronously to the rising edge of the reference clock. RX_ERR is asserted high for every reception error, including errors during Idle. Unused in RGMII mode.
PWDN/INT	21		Power Down(Default)/Interrupt: The default function of this pin is power down. Register access is required to configure this pin as an interrupt. In power down function, an active low signal on this pin places the device in power down mode. When this pin is configured as an interrupt pin, this pin is asserted low when an interrupt condition occurs. The pin has an open-drain output with a weak internal pullup (9.5 k Ω). Some applications may require an external PU resistor.
TX_CLK	22		MIl Transmit Clock: MIl Transmit Clock provides a 2.5-MHz reference clock for 10-Mbps speed. Unused in RMII mode. RGMII Transmit Clock: The clock is sourced from the MAC layer to the PHY. When operating at 10-Mbps speed, this clock must be 2.5-MHz.
TX_EN	23		Transmit Enable: TX_EN is presented on the rising edge of the TX_CLK. TX_EN indicates the presence of valid data inputs on TX_D[3:0] in MII mode and on TX_D[1:0] in RMII mode. TX_EN is an active high signal. RGMII Transmit Control: TX_CTRL combines transmit enable and transmit error signals. TX_EN is presented on the rising edge of TX_CLK and TX_ER on the falling edge of TX_CLK.
TX_D0	24		Transmit Data: In MII mode, the transmit data nibble received from the MAC is synchronous to the rising edge of TX_CLK. In RMII Master mode, TX_D[0,1] are synchronous to CLKOUT50M output of the device In RMII Slave mode, TX_D[0,1] are synchronous to rising edge of Ref clock
TX_D1	25		
TX_D2	26		
TX_D3	27		
LED_2/ TX_ER	28	Strap	This pin acts as LED_2 by default. It can be configured as GPIO or TX_ER as well. The LED is ON when link is negotiated for 10M (short reach). LED remains OFF otherwise.
LED_0	29	Strap	LED : Activity Indication LED indicates transmit and receive activity in addition to the status of the link. The LED is ON when link is good. The LED blinks when the transmitter or receiver is active. This pin can also act as GPIO using register configuration.
CLKOUT/ LED_1	30		This pin provides Reference CLKOUT of 25 MHz as default to clock other module on the board. The pin can be configured to act as LED_1 using strap or register configuration. The LED is ON when link is negotiated for 10M (long reach). The LED remains OFF otherwise. When configured for CLK_OUT, reference clock is not affected by reset.
RST_N	31		RST_N: This pin is an active low reset input. Asserting this pin low for at least 25 μ s will force a reset process to occur. Initiation of reset causes strap pins to be re-scanned and resets all the internal registers of the PHY to default value.
GPIO1	32	Strap	General Purpose Input or Output.

Table 4-1. Internal PU/PD in various states						
Pin #	Pin Name	Reset State	Active State (MII Mode)	Active State (RMII Master Mode)	Active State (RMII Slave Mode)	Active State (RGMII Mode)
1	DVDD	A	A	A	A	A
2	CEXT	A	A	A	A	A
3	VDDA	A	A	A	A	A
4	TX+	A	A	A	A	A
5	RX+	A	A	A	A	A
6	RX-	A	A	A	A	A
7	TX-	A	A	A	A	A
8	GPIO2	I,PD	I,PD	I,PD	I,PD	I,PD
9	XO	A	A	A	A	A
10	XI/50MHzIn	A	A	A	A	A
11	MDIO	IO	IO	IO	IO	IO
12	MDC	I	I	I	I	I
13	RX_D3	I,PD	O,Hi-Z	I,PD	I,PD	O,Hi-Z
14	RX_D2	I,PD	O,Hi-Z	I,PD	I,PD	O,Hi-Z
15	RX_D1	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
16	RX_D0	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
17	VDDIO	A	A	A	A	A
18	RX_DV/ CRS_DV	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
19	RX_CLK/ 50MHz)RMII_M	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
20	RX_ER	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	I,PD
21	PWDN/INT	I,PU-9.5K Ω /OPEN DRAIN	I,PU-9.5K Ω /OPEN DRAIN	I,PU-9.5K Ω /OPEN DRAIN	I,PU-9.5K Ω /OPEN DRAIN	I,PU-9.5K Ω /OPEN DRAIN
22	TX_CLK	I,PD	O,Hi-Z	I,PD	I,PD	I,PD
23	TX_EN	I,PD	I,PD	I,PD	I,PD	I,PD
24	TX_D0	I,PD	I,PD	I,PD	I,PD	I,PD
25	TX_D1	I,PD	I,PD	I,PD	I,PD	I,PD
26	TX_D2	I,PD	I,PD	I,PD	I,PD	I,PD
27	TX_D3	I,PD	I,PD	I,PD	I,PD	I,PD
28	LED_2/ TX_ER	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
29	LED_0	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
30	CLKOUT/ LED_1	I,PD(Only at POR)	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z
31	RST_N	I,PU	I,PU	I,PU	I,PU	I,PU
32	GPIO1	I,PD	O,Hi-Z	O,Hi-Z	O,Hi-Z	O,Hi-Z

The definitions below define the functionality of the I/O cells for each pin. (a) Type: I - Input (b) Type: O - Output (c) Type: I/O - Input/Output (d) Type OD - Open Drain (e) Type: PD, PU - Internal Pulldown/Pullup (g) Type Hi-Z : floating (h) Type:A - Analog

PHY offers interal PU or PD resistor for the default strap configuration and eliminates need for external resistor. External resistor for strap is needed only when default configuratoin needs to be changed.

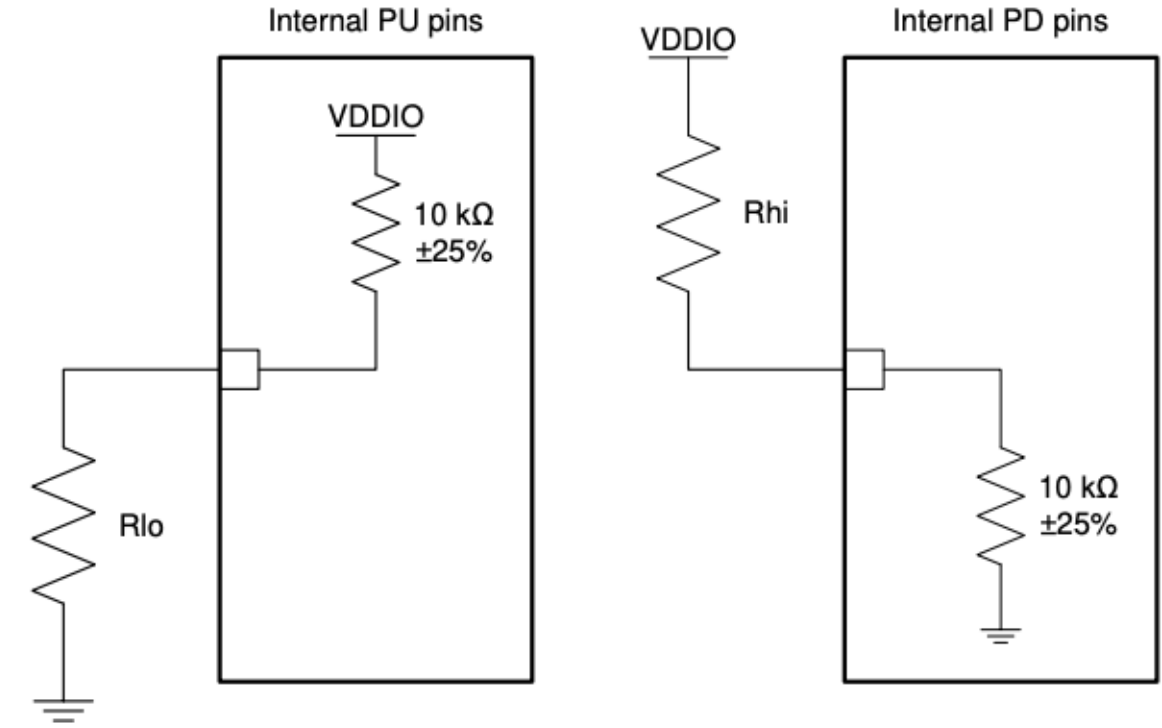


Figure 6-10. Strap Circuit

Table 6-7. 2-Level Strap Resistor Ratio

MODE	IDEAL RESISTORS	
	Rhi (k Ω)	Rlo (k Ω)
0	OPEN	2.49
1	2.49	OPEN

6.4.1.1 Straps for PHY Address

Table 6-8. PHY Address Strap Table

PIN NAME	STRAP NAME	PIN #	DEFAULT	
GPIO1	Strap9	32	0	PHY_ADD0
				MODE 0
				MODE 1
RX_ERR	Strap6	20	0	PHY_ADD1
				MODE 0
				MODE 1
RX_D0	Strap4	16	0	PHY_ADD2
				MODE 0
				MODE 1
RX_D3	Strap1	13	0	PHY_ADD3
				MODE 0
				MODE 1

PHY Address strap is 4 bit strap on pin 13, 16, 20 and 32. It shall be read as [3:2:1:0] respectively. Default PHY Address is 0000.

Table 6-9. Reach Selection Strap

PIN NAME	STRAP NAME	PIN #	DEFAULT		
LED_2	Strap7	28	0	0	This Strap defines the voltage level requested by PHY during auto negotiation. It is reflected in bit 12 of 0x20E. While using Force mode for Linkup, the strap controls the output voltage and reflects in bit 12 of 0x18F6 0 : 2.4V & 1-V p2p
				1	1: 1-V p2p

Table 6-10. MAC Mode Strap Table

PIN NAME	STRAP NAME	PIN #	DEFAULT	Strap8	Strap3	
RX_D1	Strap3	15	0	0	0	MIl (default)
				0	1	RMII Master
LED_0	Strap8	29	0	1	0	RGMII
				1	1	RMII Slave

Table 6-11. RMII MAC Mode Strap Table

PIN NAME	STRAP NAME	PIN #	DEFAULT		
RX_D2	Strap2	14	0	0	Pin 18 is configured as CRS_DV (Default, for Media Conversion Mode)
				1	Pin 18 is configured as RX_DV (For RMII Repeater or Media Conversion Mode)

Table 6-12. Terminations Selection

PIN NAME	STRAP NAME	PIN #	DEFAULT		
GPIO2	Strap10	8	Mandatory PU/PD	0	Receiver with tapping at 50 Ω (Recommended)
				1	Receiver tapping at < 40 Ω

Table 6-13. Clockout/LED_1

PIN NAME	STRAP NAME	PIN #	DEFAULT		
RX_DV/CRS_DV	Strap5	18	0	0	Clockout 25 M(default)
				1	LED1